High Voltage Battery Replacement (Truck)





12v negative battery cable

WARNING

The high voltage battery is operated by both low and high voltage, so the proper safety procedures must be followed to service the component.

Prior to Removal:

Follow the High Voltage disabling/enabling procedure

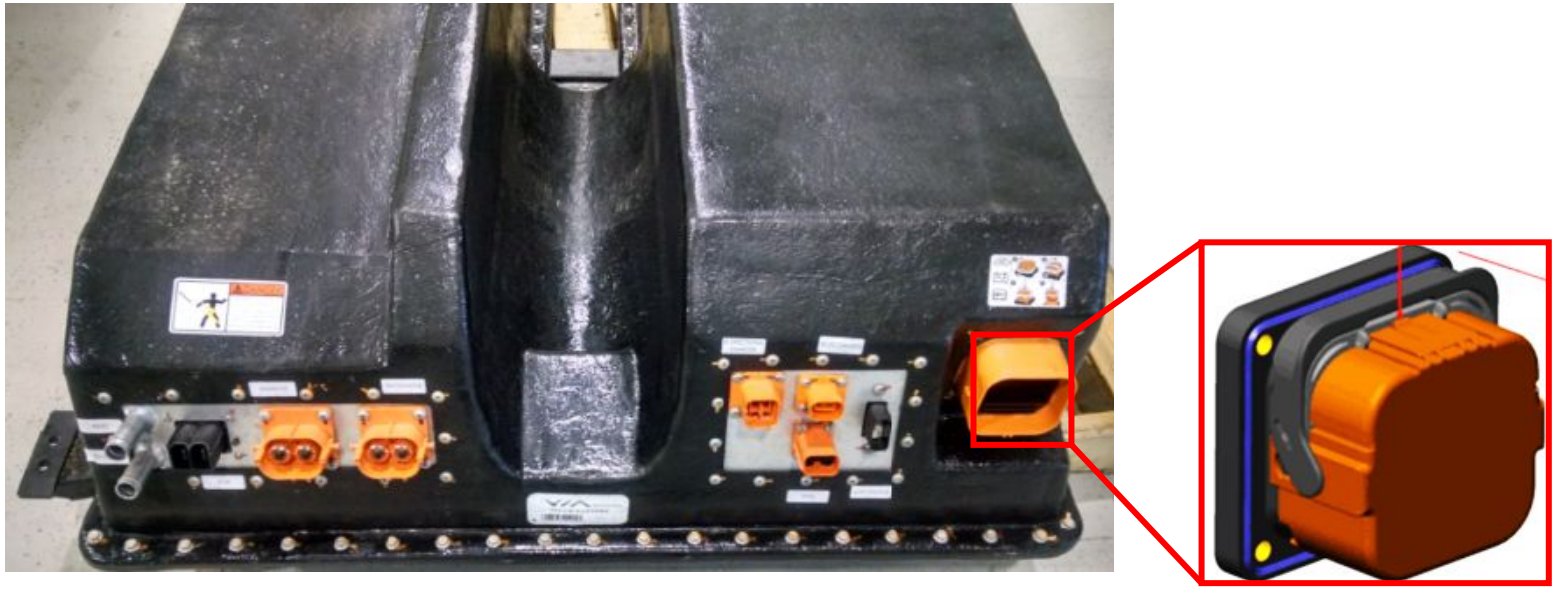
Remove the 12v negative battery cable

Make sure the vehicle is properly supported.

Raise the vehicle.

Remove the MSD located on the driver side front of the battery.

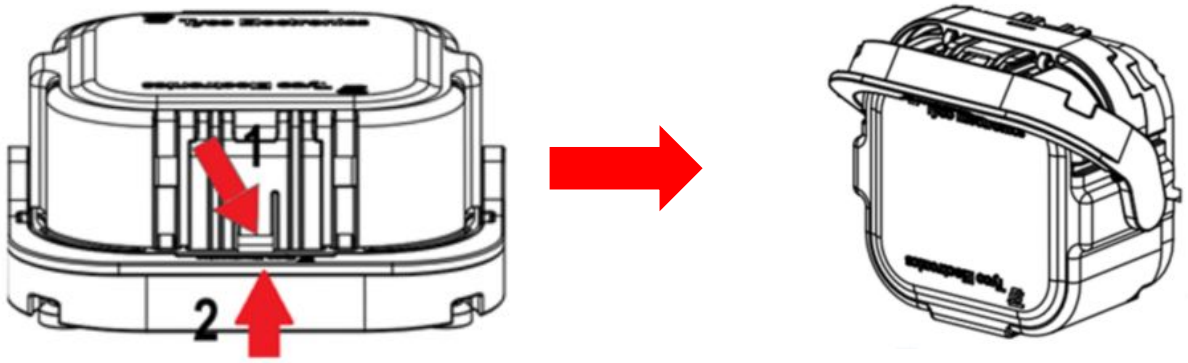
High Voltage Battery Replacement (Truck)

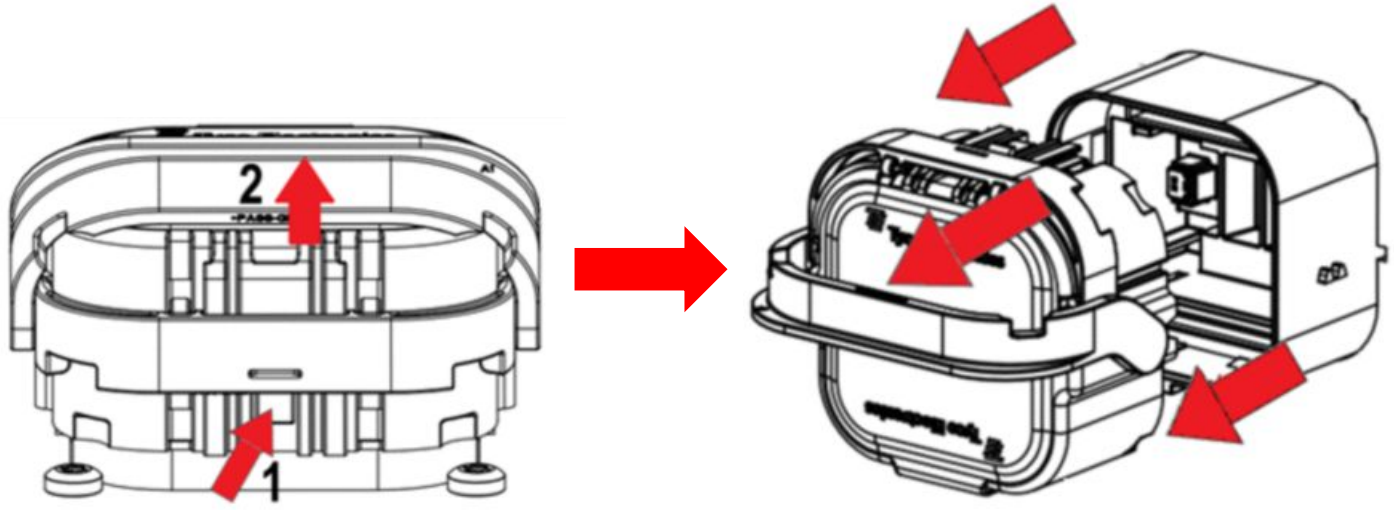


Disable the high voltage battery by removing the Manual Service Disconnect (MSD). The MSD is located on the battery pack and utilizes a two stage lever to open the HVIL circuit prior to separation of HV contacts. Removal of the MSD internally disconnects the HV side of the battery pack. Internal components of the MSD are not serviceable. In the event of part damage or failure, remove and replace the entire assembly. DO NOT attempt to separately replace single MSD components.

(See next page for removal instructions)

High Voltage Battery Replacement (Truck)





The MSD is located on the driver side front of the high voltage battery. Reach through the crossmember and release the MSD using the following procedure. Depress the latch (1) firmly. With the latch fully depressed, rotate the lever (2) upward and away from the battery pack.

Rotate the lever arm until it stops in the intermediate position at approximately 45° from the battery pack housing. Depress the latch (1) firmly. With the latch fully depressed, rotate the lever (2) upward and away from the battery pack. Continue to rotate the lever to the end of travel (90° from the battery pack) and pull straight back on the MSD to remove.

High Voltage Battery Replacement (Truck)



The High Voltage pack has a weight of 900lbs. Using equipment with insufficient capacity or stability may cause a falling or tipping hazard and should be avoided.

We recommend using a NORCO lift for this procedure.

Caution

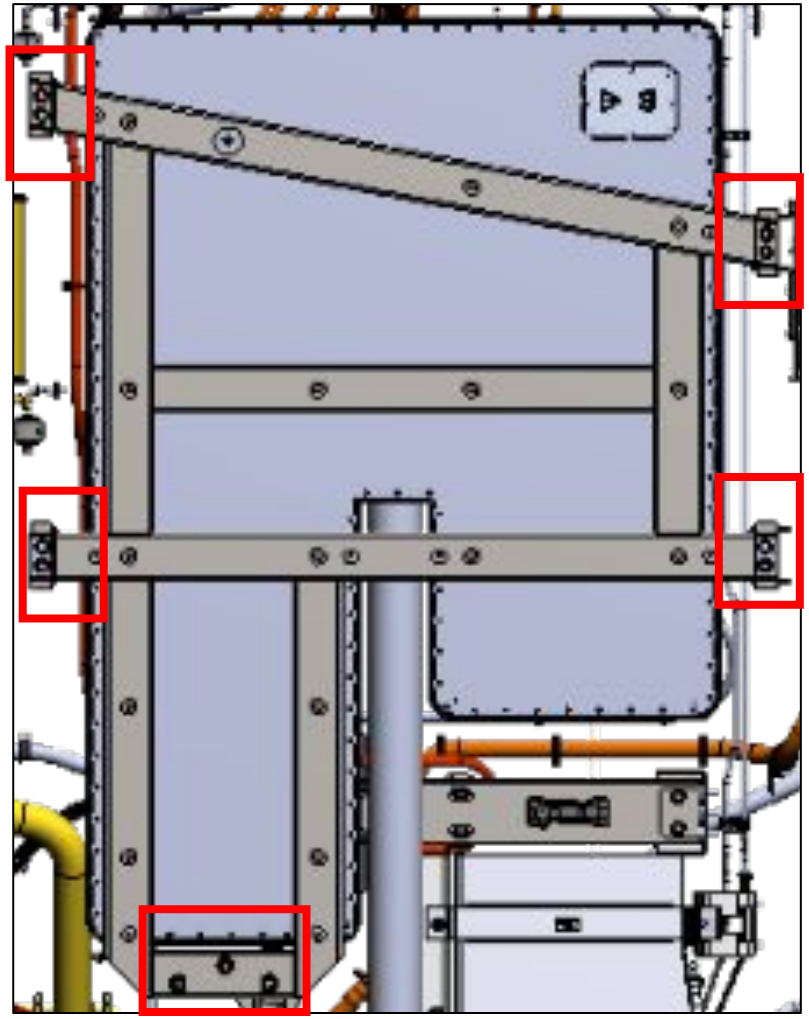
EXTREME CAUTION SHOULD BE USED WHILE LIFTING OR MOVING ASSEMBLY.

SEVERE INJURY OR DEATH MAY OCCURE IF ASSEMBLY SHIFTS FROM LIFT.

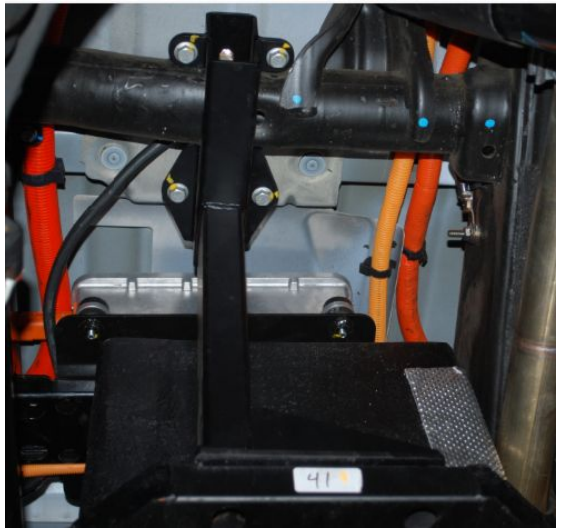
High Voltage Battery Replacement (Truck)

Be sure to center the lift under the battery and remove the 11, 15mm bolts.





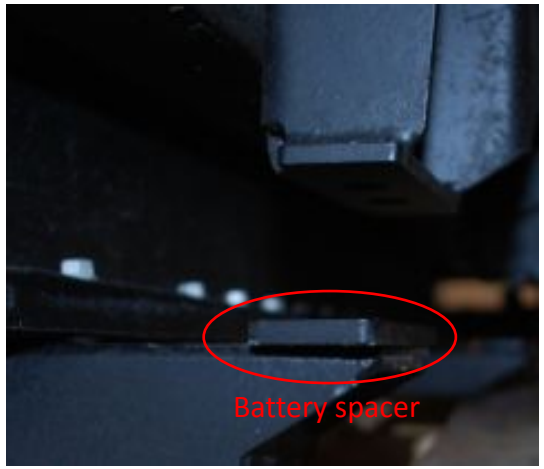
High Voltage Battery Replacement (Truck)



Rear battery support







As you begin to lower the battery rotate the rear battery support towards the rear axle to provide clearance and allow for the battery to be moved backwards during removal.

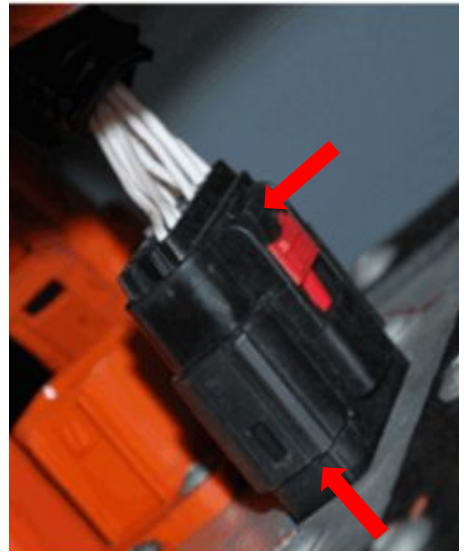
Lower the battery enough to gain access to the HV connectors and coolant lines. Approximately 2.5 inches

If the vehicle has spacers at the mounting points retain them for reinstallation.

High Voltage Battery Replacement (Truck)

The sixteen pin low voltage connector is removed by pulling rearward on the red tab and then pushing down to release it

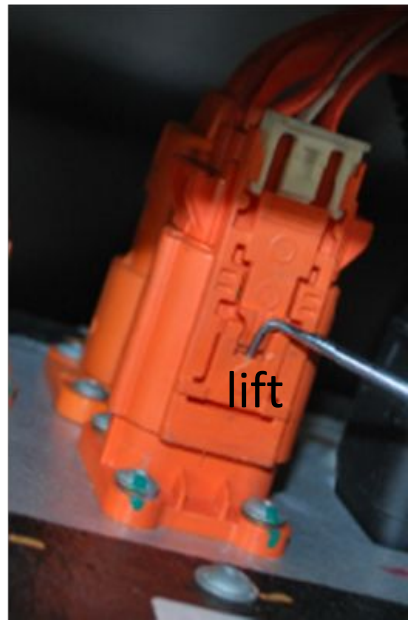




To remove the HVAC connector, slide back the white wedge lock, push down on the first lock and slide back until it stops then lift the orange tab to release







High Voltage Battery Replacement (Truck)





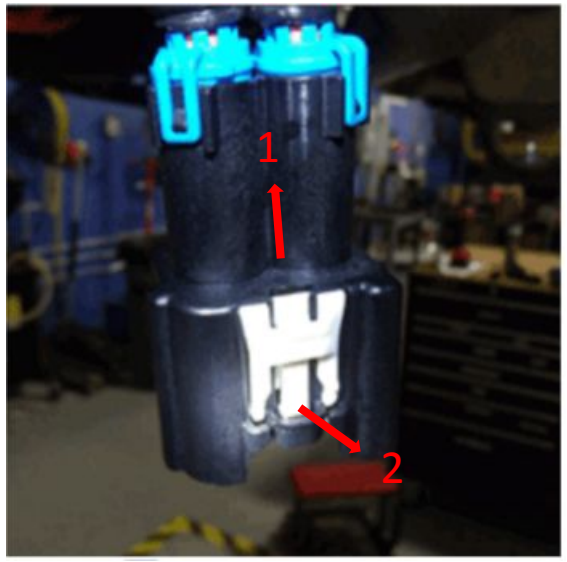
The bi-directional high voltage lead can be removed by pulling rearward on the white tab then pressing down on it.







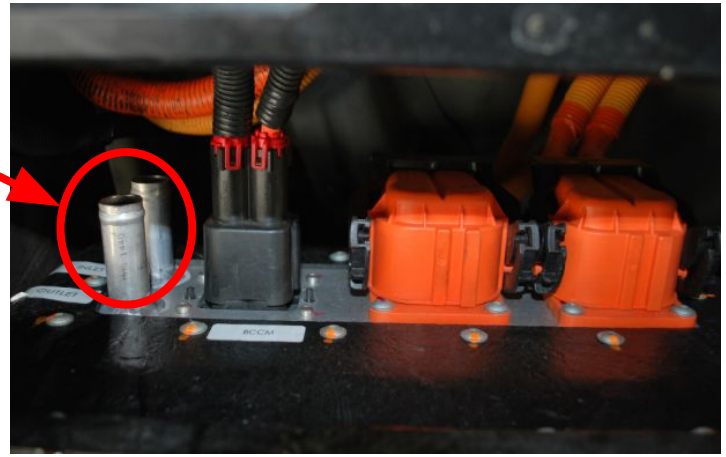
Remove the black BCCM 12 V connector from the battery by pulling back on the white tab to unlock it then lifting on the black tab to remove it.



The DCDC high voltage lead has a three stage connector, first pull back on the red tab to unlock the connector. Second push the orange tab in and pull back until it stops. Third push in on the black tab, pull back to remove the rest of the way.

High Voltage Battery Replacement (Truck)

Coolant fittings





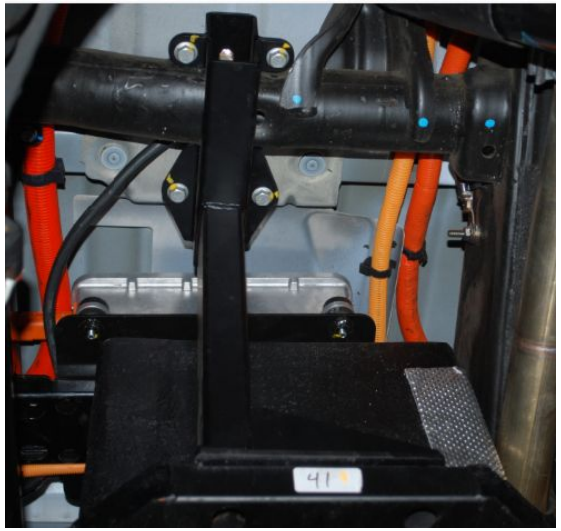
Clamp and remove the two coolant lines. Place a drain pan underneath to contain any leakage when removed.



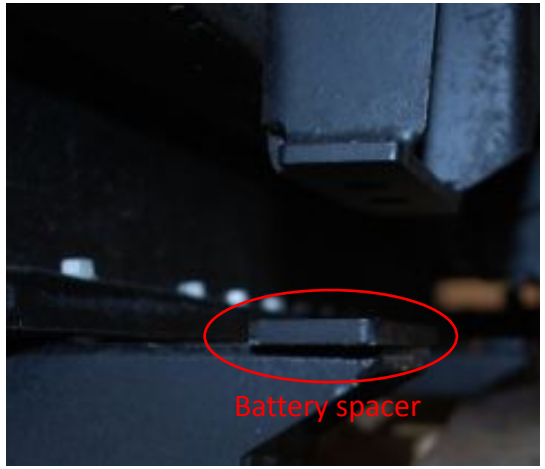


Remove generator and traction inverter connectors. To release pull back evenly on the black handle and rotate down to release the connector. If it does not come easily do not force it as the locking tabs may break.

High Voltage Battery Replacement (Truck)

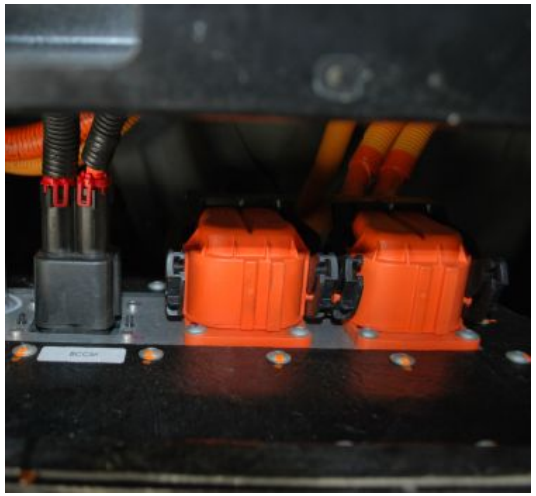


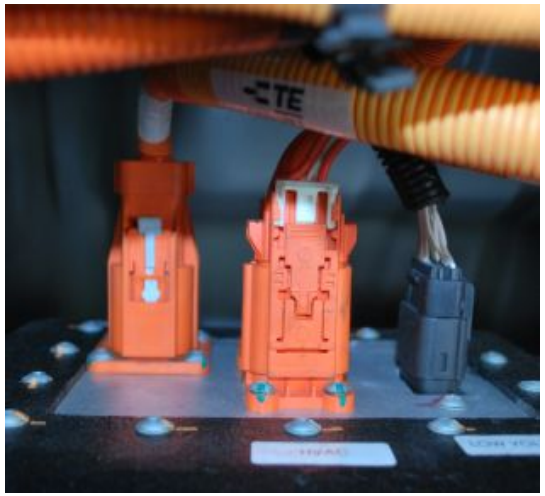




Once the HV connectors and coolant lines have been removed, slowly lower the battery, making sure it is steady on the lift, and remove from vehicle.

High Voltage Battery Replacement (Truck)









Begin replacing the battery connections. Make sure the connections are fully seated and all locks are in place.

Install the two coolant hoses in the proper location and remove the clamps

Raise the HV battery into position watching for interference. Coat the bolts with blue Loctite 242 before installing.

Torque the battery to frame abutments to 35 ft lb.

Remove the battery lift equipment.

Reconnect the high voltage system by installing the MSD.

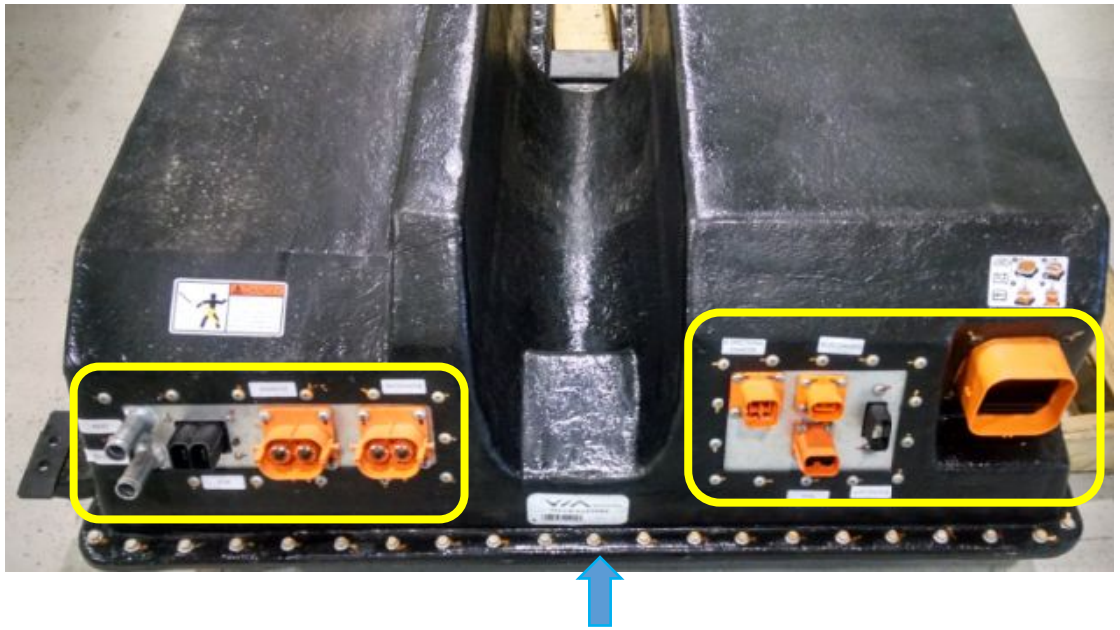
Lower the vehicle, reconnect the negative battery cable.

Refill the coolant system (see fluid filling document)

Test vehicle for proper operation

High Voltage Battery Replacement (Truck)

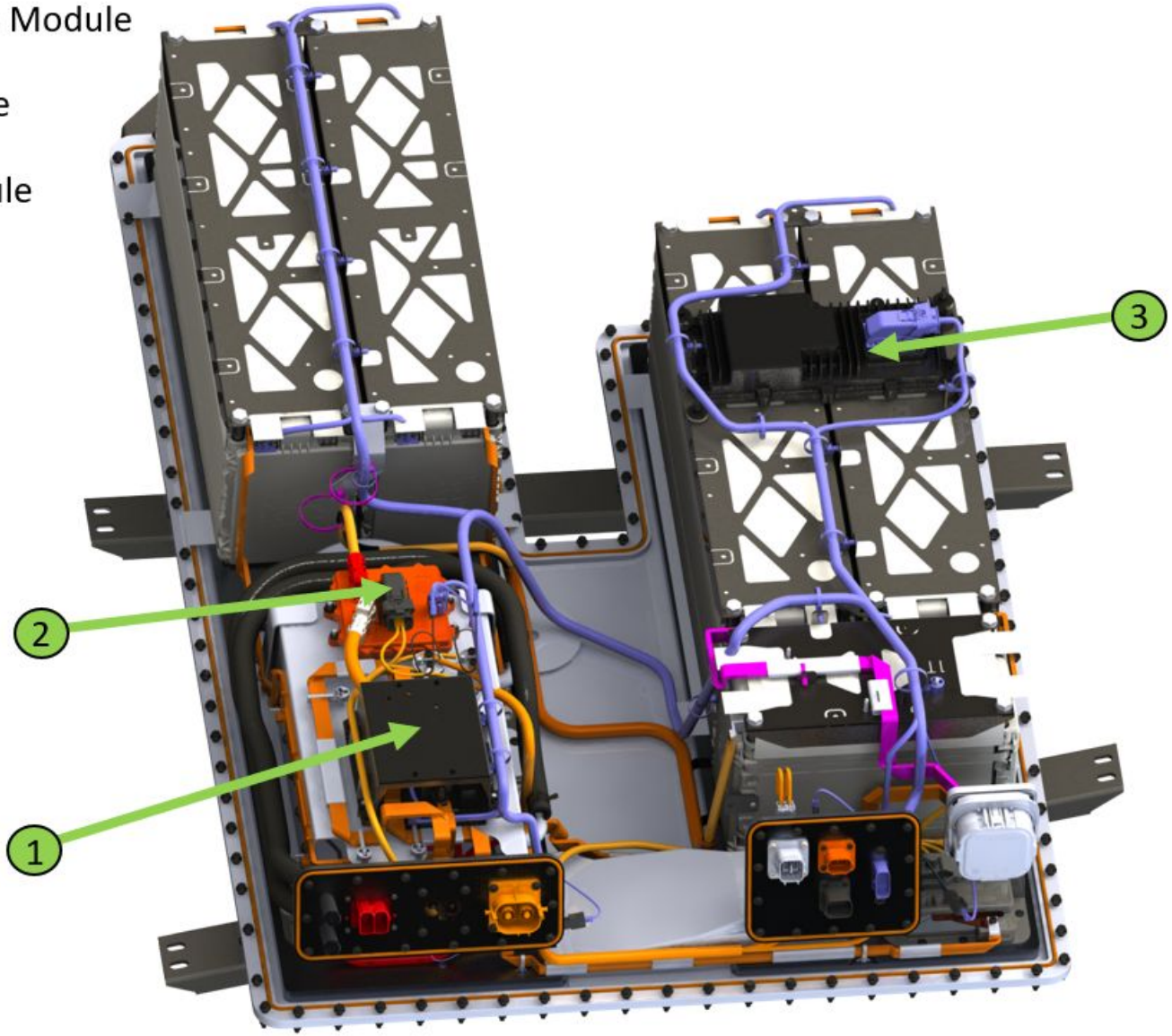
- Remove connector plate mounting bolts (Yellow) using the correct Allen socket. (It is usually necessary to lightly strike the bolt heads to break the fastener seal to the case)
- Remove battery case bolts (Blue) using two 10mm. Carefully lift upper case at the rear of the pack to clear the battery modules. Then move forward to clear the connector plates.
- Clean all seals and seal surfaces before reassembly.



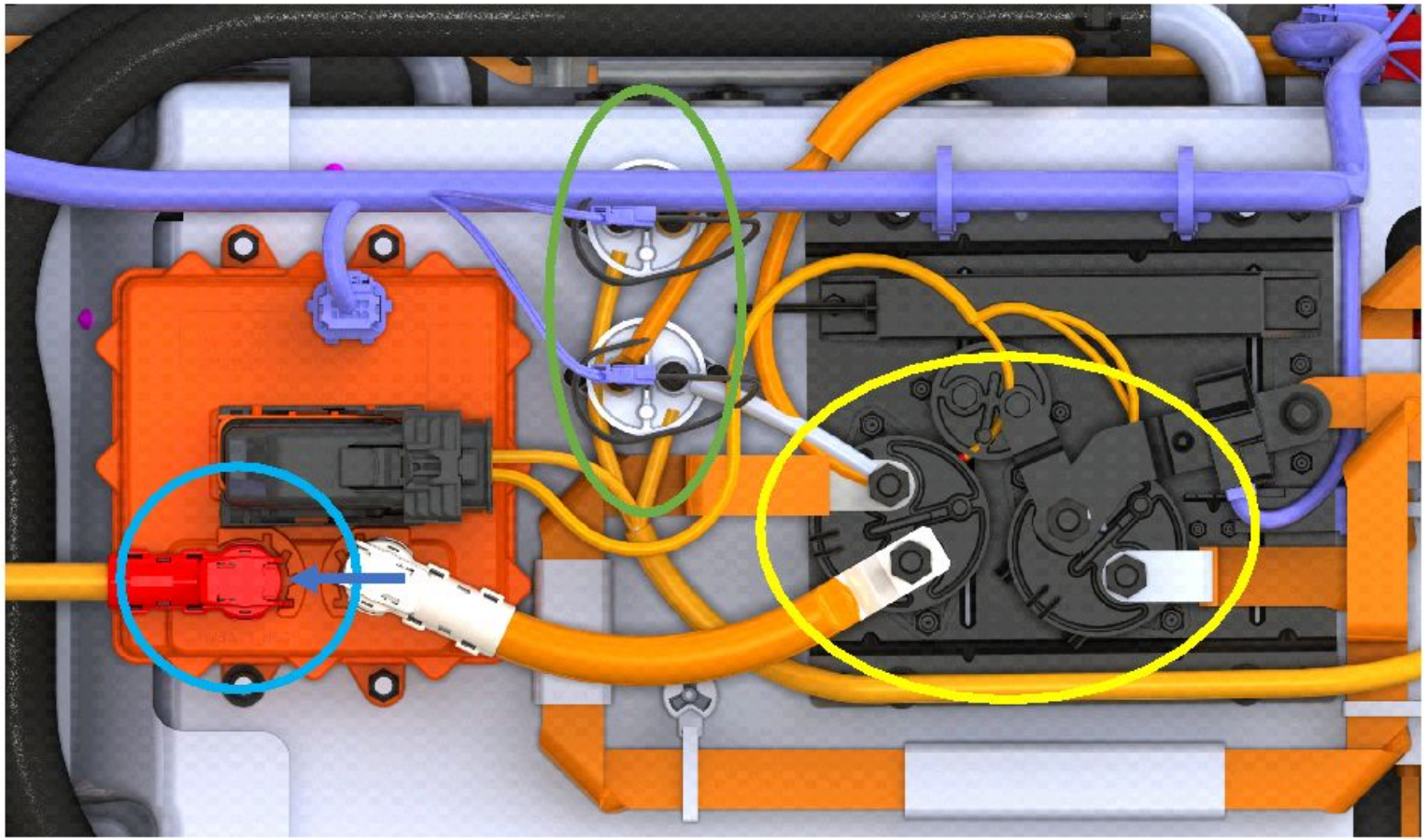
- Torque case bolts to 11Nm.
- Torque Connector plates to 7Nm.

High Voltage Battery Replacement (Truck)

1. EDM: Electrical Distribution Module
2. CSM: Current Sense Module
3. BCM: Battery Control Module

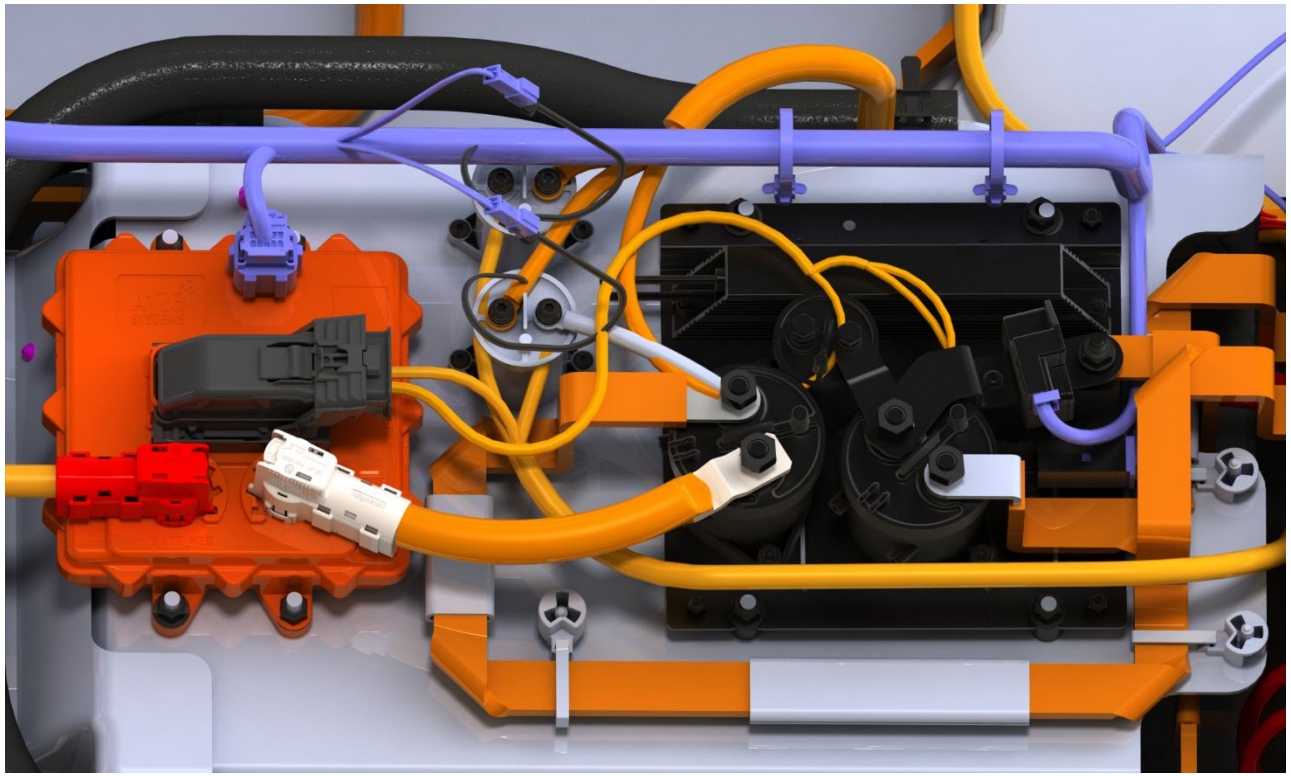


High Voltage Battery Replacement (Truck)



High Voltage Battery Replacement (Truck)

- Remove harness clips from EDM cover using a plastic clip tool.
- Remove four EDM cover nuts and washers. Remove cover
- Remove main contactor cable nuts and washers and disconnect small harness connectors. (Note all locations)
- Remove four EDM mounting hardware and remove EDM. Revers order to install new EDM.
- Torque EDM mounting and Main contactor nuts to 11Nm.
- Cover- 4Nm



- Remove Charge contactor wire connection bolts. (Note locations)
- Disconnect small harness connecters. (Note locations)
- Remove Charge contactor mounting bolts. And replace contactors. Torque to 4Nm.

High Voltage Battery Replacement (Truck)

While reassembling the battery case, be sure to align all connector plates before lowering the rear of the cover over the modules.
Start all cover bolts before tightening any others.
Tighten bolts from the center and work outward.